

SAINT JOHN, NB, Canada

WMO: 716090

Lat: 45.318N

Lon: 65.886W

Elev: 357

StdP: 14.51

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-2.7	-19.2	1.9	-5.4	-14.4	2.5	-1.5	34.7	38.2	30.0	29.8	7.5	0	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	79.3	66.1	76.5	64.3	73.8	62.9	68.9	75.4	67.0	72.8	65.3	70.6	10.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	99.2	71.4	64.8	93.2	69.4	63.3	88.1	67.7	33.3	75.8	31.7	73.0	30.4	70.6	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.9	24.3	21.4	DB	-14.1	86.1	4.5	2.5	-17.3	87.9	-20.0	89.4	-22.5	90.8	-25.8	92.7	(4)
(5)			WB	-14.6	72.5	4.5	1.7	-17.8	73.7	-20.5	74.7	-23.0	75.7	-26.2	76.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	18.6	20.3	28.1	39.0	49.2	57.6	63.4	63.1	56.5	46.6	36.7	26.3
	DBStd	17.41	11.78	10.65	8.97	6.09	5.33	5.23	3.85	4.01	6.15	6.83	8.62	10.13
	HDD50	4208	975	832	679	332	80	5	0	0	14	151	405	735
	HDD65	8371	1440	1252	1144	780	489	227	76	83	260	572	848	1200
	CDD50	1376	0	0	0	2	55	235	417	406	210	44	7	0
	CDD65	64	0	0	0	0	0	6	28	24	6	0	0	0
	CDH74	566	0	0	0	0	10	86	242	188	38	2	0	0
	CDH80	73	0	0	0	0	1	11	32	22	7	0	0	0
Wind	WSAvg	9.7	10.8	10.9	11.2	10.9	9.6	8.7	7.5	7.3	8.2	9.7	10.3	10.8
Precipitation	PrecAvg	49.7	4.3	3.9	4.1	3.6	3.9	3.8	3.5	3.0	4.4	5.1	5.1	5.1
	PrecMax	69.7	7.1	8.5	8.1	6.7	8.9	8.9	7.1	6.6	9.5	9.4	8.7	10.6
	PrecMin	30.1	1.5	1.2	0.7	1.1	1.5	0.8	1.0	0.4	1.2	1.1	1.7	1.3
	PrecStd	8.6	1.6	1.7	1.7	1.5	1.7	2.2	1.8	1.7	2.4	2.2	1.9	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.5	48.1	53.7	62.3	74.5	81.6	84.0	83.0	78.4	69.2	61.2	53.3
		MCWB	49.5	45.6	44.0	50.3	59.0	65.9	68.8	68.2	65.9	59.9	58.1	51.8
	2%	DB	45.1	42.4	47.4	57.0	68.5	76.2	79.3	78.5	73.2	63.8	57.3	49.2
		MCWB	44.1	40.3	42.1	48.0	55.7	62.6	66.3	66.0	63.7	58.4	55.3	48.0
	5%	DB	40.6	38.8	43.3	53.3	64.3	72.3	76.6	75.7	69.8	60.4	53.7	45.4
		MCWB	39.2	36.7	39.1	45.8	52.9	60.3	64.7	64.2	61.7	56.1	51.5	43.6
	10%	DB	36.3	36.1	40.0	49.9	60.4	68.9	73.6	73.1	66.7	57.8	50.5	41.3
		MCWB	34.4	33.9	36.5	43.3	51.6	58.5	63.4	63.1	60.4	54.0	48.2	39.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	46.4	49.6	54.9	62.5	68.0	71.7	71.5	69.3	63.6	59.4	52.2
		MCDB	50.7	47.7	51.8	57.3	70.4	77.6	80.3	79.0	74.9	66.3	60.6	52.8
	2%	WB	44.3	40.6	43.5	50.8	58.6	65.2	69.0	68.8	66.3	60.3	55.7	48.4
		MCDB	45.0	41.8	45.7	54.3	64.3	72.8	76.1	74.7	69.9	62.4	56.8	49.3
	5%	WB	39.4	37.1	39.9	47.4	55.7	62.7	67.0	66.8	64.0	57.5	52.0	44.3
		MCDB	40.3	38.5	42.4	51.2	61.3	68.9	73.3	72.1	67.5	59.4	53.2	45.4
	10%	WB	34.5	34.1	36.8	44.5	53.0	60.6	65.2	65.0	62.0	54.7	48.3	39.5
		MCDB	36.1	36.0	39.6	48.7	58.2	65.9	70.8	70.0	65.3	56.9	50.1	41.1
Mean Daily Temperature Range	5% DB	MDBR	17.8	17.9	16.1	16.4	18.4	18.8	18.2	18.6	18.2	16.3	14.5	15.2
		MCDBR	22.3	16.6	19.2	23.0	26.4	25.5	24.6	24.3	21.7	19.7	17.0	17.7
		MCWBR	23.2	15.7	15.1	15.3	15.7	14.2	13.3	13.4	13.4	14.5	16.2	18.1
	5% WB	MCDBR	22.2	16.6	16.7	18.6	20.8	20.1	19.5	18.5	16.3	15.6	15.4	17.5
		MCWBR	23.1	16.5	15.2	14.8	14.1	12.9	12.1	11.7	11.7	13.6	15.9	18.5
Clear-Sky Solar Irradiance	taub		0.284	0.290	0.319	0.350	0.373	0.410	0.431	0.405	0.374	0.331	0.309	0.295
	taud		2.452	2.442	2.436	2.398	2.381	2.309	2.276	2.356	2.410	2.527	2.533	2.459
	Ebn at Noon		266	286	291	289	284	272	265	268	268	267	254	248
	Edh at Noon		22	27	31	35	37	40	41	36	32	25	21	20
All-Sky Solar Radiation	RadAvg		460	733	1101	1374	1584	1720	1759	1572	1266	803	458	350
	RadStd		42	57	73	128	176	147	128	104	91	82	50	29

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (31 neighbors)	N/A	N/A	N/A	N/A	+1.33	+1.85	N/A	N/A	+76	N/A
	N/A	N/A	N/A	N/A	+0.61	+0.83	N/A	N/A	+97	N/A

Nomenclature: See separate page